Advanced Digital Design
130323
Simulation with Testbench (Assignment 3)
Date: Tuesday, November 19, 2024

Submission:	**By not more than 2 people together**
Deadline:	**Tuesday, December 3, 2024 (17:15) – via moodle**
According to JCT's policy, if you violate any submission or deadline criteria you disqualify your assignment	

A. Scope:

The purpose of this exercise is to introduce you to the concept of using a VHDL testbench in Modelsim.

In the previous assignment, we have learned how to toggle signals by forcing their values within ModelSim. We have further learned how to automate the simulation by using ModelSim's "*.do" scripts. However, ModelSim's internal commands, whether manual or automated are not enough. They are specific to ModelSim (i.e., not compatible with internal commands of other VHDL simulators) and sometimes have a limited capability when compared to the capabilities of the VHDL language.

Therefore a (V)HDL testbench, that utilize the capabilities of the (V)HDL language is required. The testbench plays the role of an external board that drives the inputs of the design and accepts the design outputs, according to a given ATP.

(V)HDL testbench is the industry standard for driving a logic simulation !

B. Preparations:

1. Create a directory for running the lab, as instructed in the previous assignment.
2. Copy **nnn.vhd** and **tb_nnn.vhd** to the new directory (Note: the **Add_Me1.vhd** and **Add_Me2.vhd** files are only for the experiment in section E)

C. The concept of using a testbench:

Figure 1 illustrates the concept of using a testbench. The design (nnn.vhd) and testbench (tb_nnn.vhd) used in the assignment are given as an example.

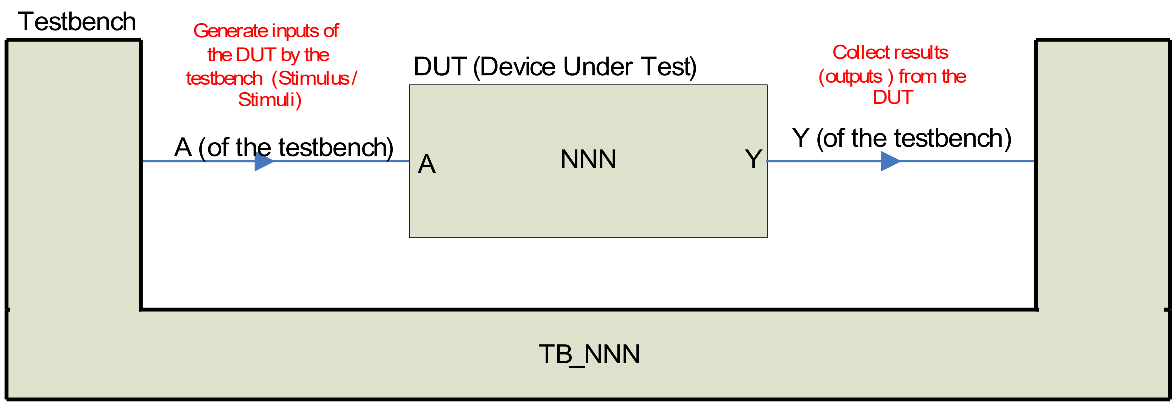

Figure 1: The concept of using a (V)HDL testbench

As illustrated, the “device under test” is driven by signals from the testbench. Output signals of the “device under test” are being monitored by viewing them in ModelSim's waveform window.

This concept is achieved by using the following constructs of the VHDL language, as given in figure 2:

```
library IEEE ;
  use IEEE .std_logic_1164 .all ;

entity TB _NNN is
end TB _NNN;

architecture YYY of TB _NNN is

  component NNN
  port
  (
    A : in  std_logic ;
    Y : out std_logic
  ) ;
  end component ;

  signal A _in_the_tb, Y _in_the_tb: std_logic ;

begin

  DUT : NNN
  port map
  (
    A => A_in_the_tb,
    Y => Y_in_the_tb
  ) ;


  A_in_the_tb <= '1',
                 '0' after  50 ns,
                 '1' after 100 ns,
                 '0' after 150 ns,
                 '1' after 200 ns,
                 '0' after 300 ns;

end YYY ;
```

Entity for the testbench (mostly, an empty entity)

Component declaration: "Copy & Paste" of the "Device Under Test" (DUT) entity, but using the word component and not using "is".

Signals for the testbench

Component Instantiation: Put the "Device Under Test" and connect the testbench' signals to the DUT pins.

Toggle the "Device Under Test" inputs (stimulus/stimuli) according to an ATP.

Figure 2: VHDL - basics of writing a testbench (tb_nnn.vhd)

Open both the design (nnn.vhd) and the testbench (tb_nnn.vhd) for reading. Make sure that you really understand the contents of both files !

D. Step by step tutorial (and exercise):

1. Start ModelSim, change to the working directory (that you prepared in B) and create a working library by using

```
vlib work
```

2. In order to run a testbench based simulation, ModelSim must receive the VHDL files in a "bottom to top" order (also known as "bottom up") - starting from the lowest level of the hierarchy ("device under test", nnn.vhd) and ending with the testbench (the top level, tb_nnn.vhd). The ModelSim commands that you must use in the transcript window are therefore:

```
vcom nnn.vhd
vcom tb_nnn.vhd
```

3. Start simulation by using the following ModelSim command:

```
vsim -voptargs=+acc=lprn tb_nnn
```

4. Make sure that ModelSim presents the "Workspace" and the "Objects" windows, as shown in figure 3

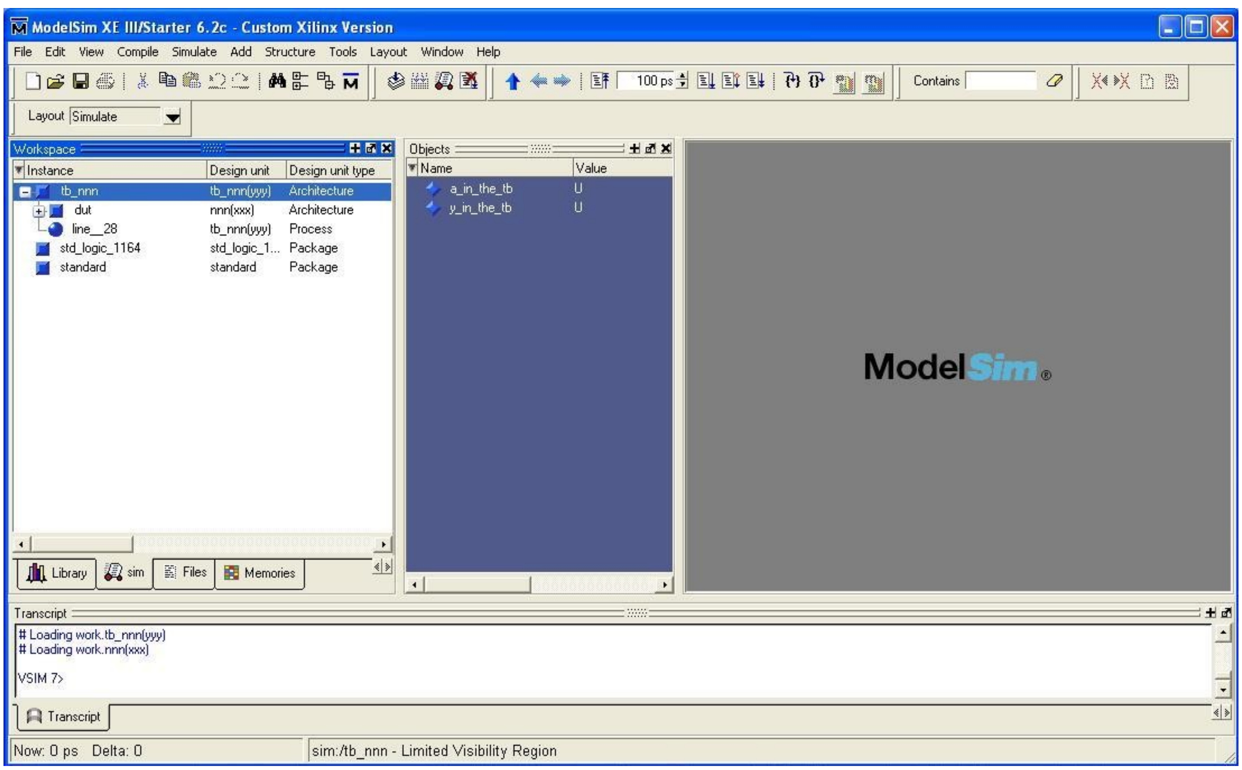

Figure 3: Workspace and Objects windows in ModelSim

If these windows are not shown by ModelSim you should use the "View" menu to show them (the menu options are different, for different ModelSim versions).

The workspace window presents the hierarchy that exists in your system (where ***system*** is the whole formation of both the design and testbench together).

5. You can further see in figure 3 that when you highlight the testbench (tb_nnn) in the “Workspace window, the Objects window presents the signals containd in the testbench (see tb_nnn.vhd for more details !).

   However, if you highlight the “device under test” signals (marked as DUT, because of the label given to it in the testbench' VHDL code) So, highlight the DUT leaf in the “Workspace” window and verify that you indeed see the signals names that were given in nnn.vhd, as seen in figure 4

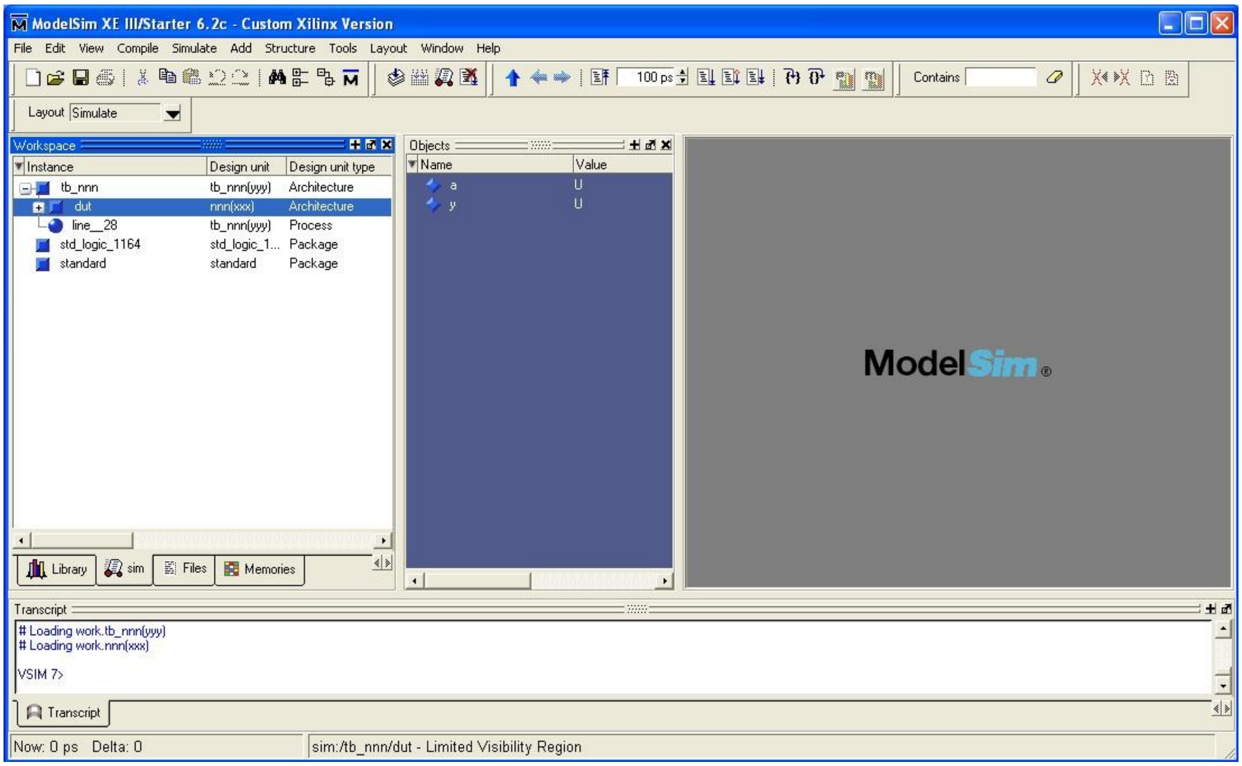

Figure 4: DUT hierarchy highlighted, showing only DUT signals in the Objects window.

6. Place the cursor in the Objects window and click on the mouse' right button. This should bring a menu where the cursor is. Select the “Add to Wave → “Signals in Design” option in order to send all the signals from all the system's hierarchies into the “Wave” window. See figure 5 for more details.

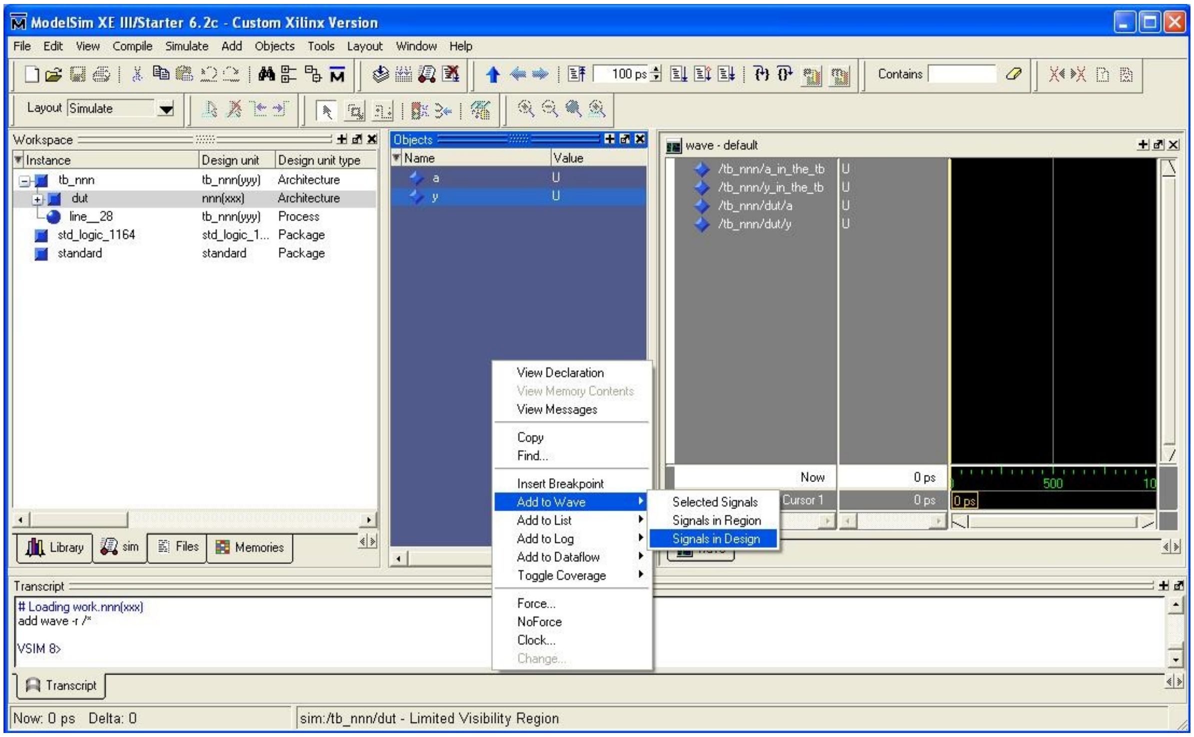

Figure 5: Add to wave --> Signals in Design

7. Run the simulation as follows:

```
run 400 ns
```

8. Select the "Full zoom" of the wavform window. Include the waveform in your report.

9. Change the testbench timing according to the times given in VHDL lecture 3, slide 19. Compile the testbench and restart the simulation. Include the new testbench and the new waveform, full zoom, in your lab report.

E. Additional Experiment

1. Open a text editor and add the contents of the **Add_Me1.vhd** and **Add_Me2.vhd** to your testbench, as follows:
   - Copy **Add_Me1.vhd** at the beginning of the file, right after the line "**use IEEE.std_logic_1164.all**;
   - Copy the contents of **Add_Me2.vhd** at the end of the file, before the line "**end YYY;**".
2. Create a new directory, copy nnn.vhd and the new tb_nnn.vhd into it, and run a new simulation.
3. Copy the contents of ModelSim's transcript window to your report
4. Add the new **tb_nnn.vhd** to your report

> Note: **Add_Me1.vhd** and **Add_Me2.vhd** contain VHDL functions and features that we didn't yet learn! We therefore have here questions and experiment of self learning and understanding.

5. Based on the simulation results, seen at the 'transcript' window, explain in your own words, to the best of your understanding, what each line do?

F. Your report must include the followings items and answers:

1. Compare, in your own words, between using a testbench instead of using ModelSim commands file (“do” file)?
2. Waveform of D.8
3. Waveform and testbench of D.9
4. Create a ModelSim commands “do” file equivalent to the original testbench. Include that “do” file in your report (this kind of “do” file replace the testbench), no need to run simulation...
5. Create a ModelSim commands “do” file equivalent to D.9 testbench. Include the “do” file in your report (this kind of “do” file replace the testbench).
6. Items required in Section E, “Additional Experiment”